

Event & Festival Module Database Specification

Complete Technical Data Dictionary, PostgreSQL Tables, Entity Mappings & Architecture

Database Engine: PostgreSQL 15+ | **Database Name:** postgres (manacommunity) | **Schema:** manacommunity
ORM: Spring Data JPA (Hibernate 6.x) | **Migrations:** Flyway V1–V58 | **Total Module Tables:** 28 Tables

Module 1: Core Events, Schedule & Venues

Main festival configurations, dynamic UPI payments, online/cash modes, program timelines and multi-venue facilities.

Table Name (PostgreSQL)	Java Entity Class	Key Columns & Functional Purpose
community_event	CommunityEvent.java	Core event configuration, title, type (festival/sports/cultural), capacity, paymentModes (UPI/Card/Cash), UPI ID, dynamic scanner QR URL, instructions, notes, contacts JSON, status.
event_venues	EventVenue.java	Physical venues, halls, mandaps, ground locations, seat capacities, gate directions, contact person and Google map coordinates.
event_program	EventProgram.java	Day-wise schedules, festival ritual timelines, stage program blocks, pandit/artist assignments, and live status.

Module 2: Registrations, Passes & Family Devotees

Unified attendee registration passes, multi-member family attendee JSON, auto-linked Gotram, and QR security code verification.

Table Name (PostgreSQL)	Java Entity Class	Key Columns & Functional Purpose
event_booking_registrations	EventBookingRegistration.java	Unified registration ledger for Pooja, Cultural, Competitions & General Passes. Stores reg_code, devotee_count, attending_devotees, members_json, gotram, payment_receipt_url, and qr_code_url.
event_registration	EventRegistration.java	Legacy event attendees ledger with confirmation timestamps and approval status.
community_event_activity_registration	ActivityRegistration.java	Cross-link association connecting general event pass holders to individual festival sub-activities.
event_family_members	EventFamilyMember.java	Registered family members profile records (name, age, gender, relationship, gotram) for one-click pass population.

Module 3: Pooja, Archana & Seva Management

Sacred ritual offerings, multi-day homams, sankalpam gotram auto-fill, pandit assignments, and capacity slot tracking.

Table Name (PostgreSQL)	Java Entity Class	Key Columns & Functional Purpose
event_pooja_sevas	PoojaSeva.java	Configured Pooja/Homam/Archana sessions, single or multi-day schedules, duration, pandit name, mandap location, item samagri list, and slot quotas.
event_pooja_types	PoojaType.java	Master catalogue of traditional pooja offerings (Maha Ganapathi Abhishekam, Ganapati Homam, Satyanarayan Puja, Laghu Rudra, etc.).

Module 4: Food & Dining / Mahaprasadam

Community meal catering, target plates, meal types (Lunch, Dinner, Mahaprasadam), and token reservation ledger.

Table Name (PostgreSQL)	Java Entity Class	Key Columns & Functional Purpose
event_lunch_dinners	LunchDinner.java	Meal schedules, catering menu items, serving venue, target plate capacity, and pricing per plate.
community_event_meal_registration	MealRegistration.java	Devotee meal passes, allocated plate counts, dietary preferences, and gate coupon scan status.

Module 5: Cultural Activities & Stage Performances

Cultural showcases, stage lineups, solo/group performance categories, age criteria, and audio-visual requirements.

Table Name (PostgreSQL)	Java Entity Class	Key Columns & Functional Purpose
event_cultural_events	CulturalEvent.java	Scheduled stage acts, performance title, duration, stage venue, coordinator lead, slots, and guidelines.
event_cultural_categories	CulturalCategory.java	Classification categories: Classical Dance (Bharatanatyam, Kuchipudi), Devotional Chanting, Drama/Skit, Instrumental Music.
event_cultural_performance_types	CulturalPerformanceType.java	Performance formats: Solo, Duet, Group (3-8 members), Mega Group (8+ members).

Module 6: Competitions & Talent Contests

Festive community contests (Drawing, Rangoli, Sloka Chanting, Quiz), age group divisions, judging metrics, and prize slots.

Table Name (PostgreSQL)	Java Entity Class	Key Columns & Functional Purpose
event_competitions	Competition.java	Competition entries, entry fees, maximum participants capacity, venue rooms, rules, and scoring rubrics.
event_competition_categories	CompetitionCategory.java	Master categories: Drawing, Elocution, Vedic Chanting, Classical Vocal, General Quiz.
event_competition_age_groups	CompetitionAgeGroup.java	Age classification tiers: Kids (5-10 yrs), Junior (11-15 yrs), Youth (16-25 yrs), Open (All ages).

Module 7: Auctions, Hundi Offerings & Sponsorships

Live celebratory auctions (Laddu / Idol auction), devotee bidding history, Hundi donations, and corporate sponsors.

Table Name (PostgreSQL)	Java Entity Class	Key Columns & Functional Purpose
event_auction_item	EventAuctionItem.java	Auction lots (e.g. Maha Prasadam Laddu, Silver Ganesh Idol), reserve base price, current winning bid, and winner user ID.
event_auction_bid	EventAuctionBid.java	Real-time bid increments, timestamp, bidder resident details, and bid status.
event_donation	EventDonation.java	Devotee voluntary offerings, general hundi contributions, payment mode, transaction reference, and 80G receipts.
community_event_sponsor	EventSponsor.java	Commercial and resident event sponsors, sponsorship tier (Platinum, Gold, Silver), pledged funds, and banner placements.

Module 8: Financial Ledger, Invoices & Budgeting

Budget allocations, actual expenses tracking, stage sound/light decorator billing, and vendor invoices.

Table Name (PostgreSQL)	Java Entity Class	Key Columns & Functional Purpose
community_event_expense	EventExpense.java	Festival expenditure ledger, department categories (Decoration, Pandit, Sound, Flowers, Prasadam), invoices, and approvals.
event_invoice	EventInvoice.java	Invoices generated for vendor procurement, event passes billing, and sponsorship billing.

Module 9: Volunteers & Operations Management

Community volunteer onboarding, department rosters (Security, Food, Mandap, Medical), and task checklists.

Table Name (PostgreSQL)	Java Entity Class	Key Columns & Functional Purpose
community_event_volunteer	EventVolunteer.java	Volunteer registrations, assigned departments, duty shifts, lead coordinator, and attendance status.
community_event_task	EventTask.java	Organizer operational checklist tasks, due dates, priority rankings, assignment leads, and completion status.

Module 10: Multi-Day Timeline & Gallery Media

Multi-day festival milestones, tagged media categories, celebration photo galleries, and S3 asset references.

Table Name (PostgreSQL)	Java Entity Class	Key Columns & Functional Purpose
event_media_day	EventMediaDay.java	Multi-day festival timeline milestones (Day 1 – Sthapana & Kalash Puja, Day 2 – Cultural Night, Day 10 – Visarjan Rally).
event_media_category	EventMediaCategory.java	Media categorization tags (Pooja Rituals, Aarti, Cultural Stage, Procession, Volunteer Team).
event_gallery_item	EventGalleryItem.java	Celebration photo & video media repository, AWS S3 image URLs, thumbnail URLs, captions, and uploader IDs.

Key Architectural Features & Design Patterns

- **Unified Pass Ledger (event_booking_registrations):** Acts as the central booking hub with composite attendee tracking (devotee_count, members_json, and gotram) for seamless family pass generation.

- **Payment Gateway Flexibility:** Supports dynamic UPI QR scanner upload, custom VPA routing, manual cash desk mode, and automatic receipt tracking via AWS S3.
- **Real-time Capacity & Slot Quotas:** Automatic capacity decrement across Pooja slots, Meal plates, and Competition entries upon registration.
- **Enterprise Multi-Tenancy & Security:** All event tables are scoped to community_id and user principal permissions with Flyway migration versioning.